Fluids in Relativity

David Neilsen

Brigham Young University

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Introduction

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Fluids in Relativity

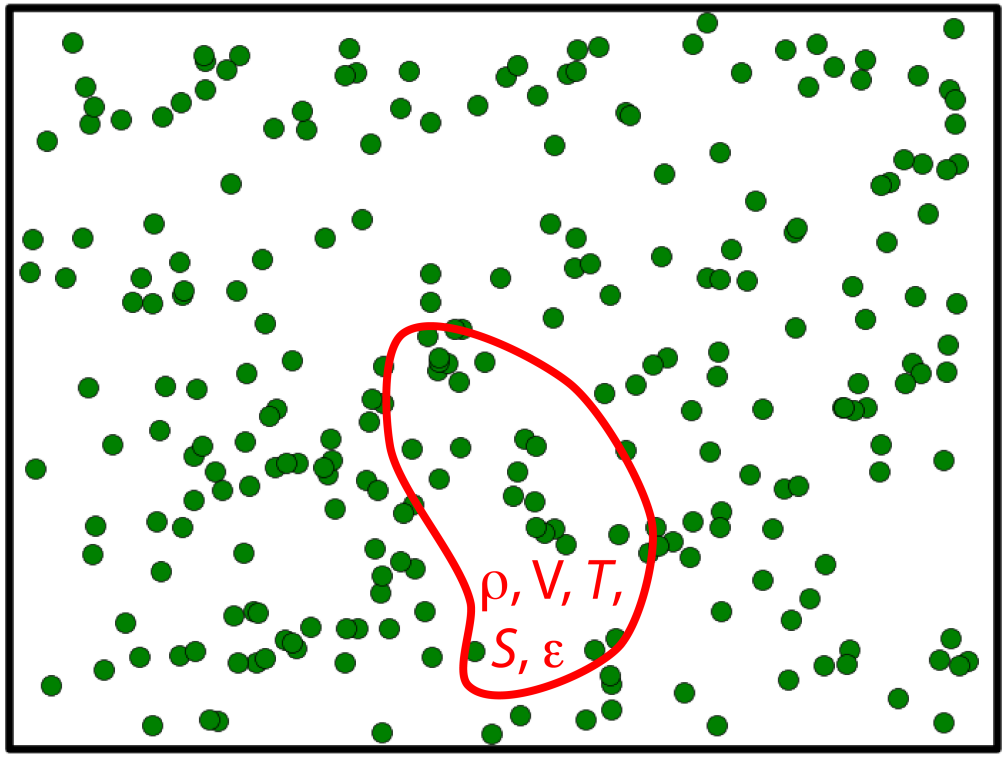
The fluid is an extremely important model in astrophysics. Fluids are used to model

- neutron stars with densities $\sim 10^{14}$ g/cm^3.
- white dwarfs with densities $\sim 10^{6}$ g/cm^3.
- accretion disks with densities $\sim 10^{-8}$ g/cm^3.
- matter and radiation in cosmology, currently $\sim 10^{-29}$ g/cm^3.
- radiation and neutrinos
- plasmas
- relativistic jets

What is a Fluid?

- A fluid is a macroscopic description of a many-bodied system of point particles out of equilibrium.
- Quantities that are not conserved during collisions decay rapidly, after a few collision times.
- Only quantities that are conserved in collisions can remain out of equilibrium.
- The hydrodynamic equations are *balance laws* or *conservation laws* for these conserved quantities, describing the long-wavelength, low-frequency behavior of the system.

What is a Fluid?



Consider an arbitrary volume inside a fluid. Inside this small volume the particles are in thermodynamic equilibrium, allowing us to ignore the 10^{23} degrees of freedom and concentrate on a few state variables to describe the system. The hydrodynamic conserved quantities include particle number, energy, and momemtum. Integral conservation laws give the equations of motion.

The *continuum hypothesis* assumes that these control volumes can be arbitrarily small, such that each control volume becomes a point. When this fails statistical mechanics must be used to describe the system.

Characterize the Fluid

- Fluid particles have a "long" mean free path compared to their size.
- Thermodynamics allows us to characterize each fluid cell or control volume using bulk variables or state variables, such as
  - n, number density
  - ρ_0, rest mass density, $\rho_0 = m_B n$
  - P, pressure
  - $\epsilon_{\rm int}$, specific internal energy
  - ϵ, total energy density
  - $\rho = \epsilon/c^2$, mass density
  - $\mathbf{u}$, average four-velocity of fluid particles
  - T, temperature

This greatly simplifies the problem because the fluid is described by only a few state variables. Note: specific internal energy is the internal energy density per rest mass density, so the total internal energy density is $\rho\epsilon_{\rm int}$.

General Balance Laws

The fluid equations of motion are conservation equations for particle number (baryons), momentum, energy. As a simple example, a conserved scalar quantity Q, and define a density q such that

$$Q = \int_{V(t)} q \, \mathrm{d}V \tag{1}$$

where the volume can change in time. In a fluid with velocity $\mathbf{v}$, the total amount of Q in $V(t)$ can change in time according to

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \int_{V(t)} \left(\frac{\mathrm{d}q}{\mathrm{d}t} + q \nabla \cdot \mathbf{v} \right) \mathrm{d}V = \int_{V(t)} \sigma_Q \, \mathrm{d}V - \oint_{S(t)} \mathbf{J}_Q \cdot \mathrm{d}\mathbf{S} \tag{2}$$

The first integral has two terms to account for the changing volume $V(t)$, and the RHS accounts for sources or sinks in $V(t)$, σ_Q, and a possible current $\mathbf{J}_Q$ through the bounding surface $S(t)$.

Balance Laws

Using integral theorems and the convective derivative

$$\frac{\mathrm{d}}{\mathrm{d}t} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \tag{3}$$

we can write equation (2) in balance law form

$$\frac{\partial q}{\partial t} = -\nabla \cdot (q\mathbf{v} + \mathbf{J}_Q) + \sigma_Q. \tag{4}$$

The quantity $q\mathbf{v}$ is the advective current, which accounts for the change of Q because the fluid is in motion.

When there are no sources ($\sigma_Q = 0$), then the equation

$$\frac{\partial q}{\partial t} + \nabla \cdot (q\mathbf{v} + \mathbf{J}_Q) = 0 \tag{5}$$

is typically called a conservation law.

Equations of State

An equation of state closes the system and contains important information about the underlying microphysical system.
Using quantum statistical mechanics, we construct a model of the many-body system. Using this model, we can define a particle number density n_{QM}, a conserved current $\mathbf{J}_{\mathrm{QM}}$, and a stress-energy tensor $\mathbf{T}_{\mathrm{QM}}$.
The fluid model is in the classical limit, and we define the observed variables as expectation values. For example, the energy density of the fluid is defined as

$$\varepsilon = \langle \mathbf{T}_{\mathrm{QM}}(\mathbf{u}, \mathbf{u}) \rangle = u_a u_b \langle {T_{\mathrm{QM}}}^{ab} \rangle \tag{6}$$

where

$$u^a \equiv \frac{1}{n} \langle {J_{\mathrm{QM}}}^a \rangle, \qquad n = \langle n_{\mathrm{QM}} \rangle \tag{7}$$

Equations of State II

The pressure is defined as

$$P = \frac{1}{3}\left(\langle T_{\mathrm{QM}}{}^{a}{}_{a}\rangle + \varepsilon\right) \tag{8}$$

At sufficiently low temperatures ε and P are functions only of n. Thus, the fluid model can describe a variety of many-body systems by changing the equation of state.

Ideal Fluid Equation of State

For simplicity we will usually assume the ideal fluid equation of state (EOS) for a gas of cold fermions. This equation of state is

$$P = (\Gamma - 1)\rho_0 \epsilon_{\text{int}} \tag{9}$$

where ρ_0 is the rest mass density, ϵ_{int} is the specific internal energy density. Γ is the $\text{adiabatic constant}$ and is the ratio of specific heats C_P/C_V. This EOS can be written in polytropic form

$$P = \kappa(S)\rho_0^{\Gamma} \tag{10}$$

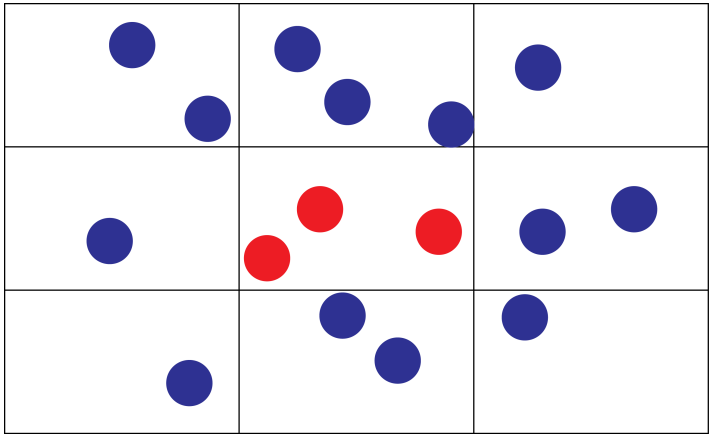
where $\kappa(S)$ is a function of the fluid entropy, S. This form of the EOS is useful for isentropic flow, as then $\kappa =$ constant.

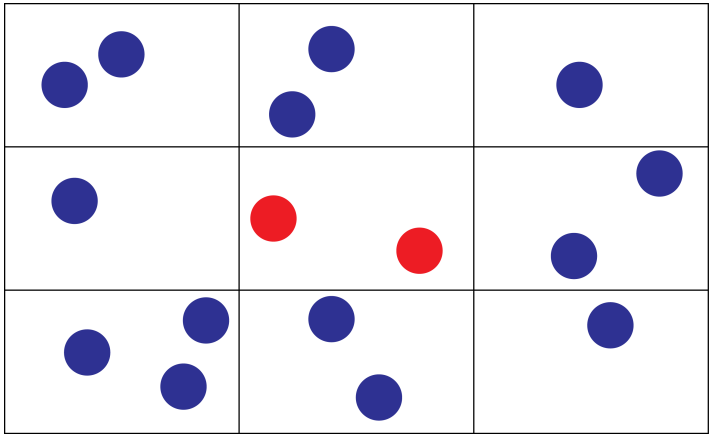
Warning

The entropy jumps across a shock, so use the polytropic form with care.

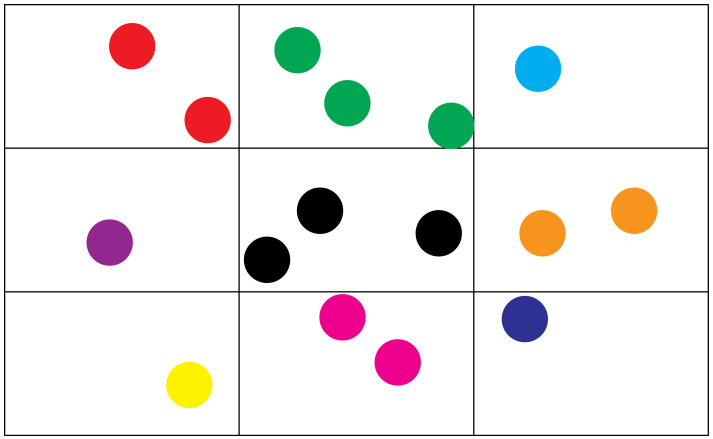
Two General Fluid Descriptions

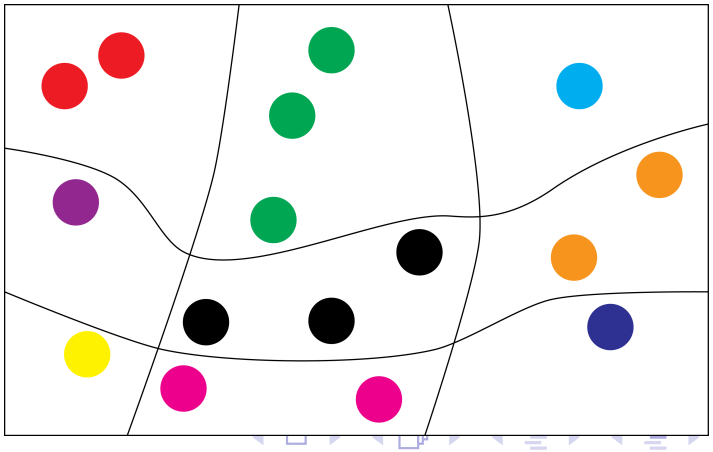
The Eulerian description of fluid dynamics, used here, uses fluid cells that are fixed to the coordinates.





In the Lagrangian description, the fluid cells follow the particles.





Fluids in Relativity: Spacetime

We assume that the continuum hypothesis holds, meaning that the state variables that describe the fluid can be defined at every point, and now work towards a description of the fluid in a general relativistic spacetime. We review quickly some parts of general relativity.

- Spacetime in relativity consists of a manifold $\mathcal{M}$ with a metric $\mathbf{g}$.
- The metric $\mathbf{g}$ has the signature $(-,+,+,+)$.
- The metric is a symmetric bilinear form

$$\mathbf{u}\cdot\mathbf{v} = \mathbf{g}(\mathbf{u},\mathbf{v}) = \mathbf{g}(\mathbf{v},\mathbf{u}) = g_{ab}u^a v^b = g_{ab}v^a u^b \tag{11}$$

- A vector $\mathbf{v}$ is timelike if $\mathbf{v}\cdot\mathbf{v} < 0$, null if $\mathbf{v}\cdot\mathbf{v} = 0$, and spacelike if $\mathbf{v}\cdot\mathbf{v} > 0$.

Particle motion

- A particle is defined by its mass $m > 0$ and its worldline $\mathcal{L}$ in spacetime. The worldline $\mathcal{L}$ is timelike.
- The proper time $\mathrm{d}\tau$ for a displacement $\mathrm{d}\mathbf{x}$ along the worldline $\mathcal{L}$ is the length given by the metric tensor

$$c\,\mathrm{d}\tau = \sqrt{-\mathbf{g}(\mathbf{x},\mathbf{x})} \tag{12}$$

- The four-velocity of the particle is the vector

$$\mathbf{u} = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\tau} \tag{13}$$

- The four-velocity is a unit vector

$$\mathbf{u}\cdot\mathbf{u} = -c^2 \tag{14}$$

The four-velocity is not defined with respect to another observer, but is instrinsic to the particle itself.

Particle motion II

- The four-acceleration is

$$\mathbf{a} = \nabla_{\mathbf{u}}\mathbf{u}. \tag{15}$$

From $\mathbf{u} \cdot \mathbf{u} = -c^2$ it is easy to show that $\mathbf{a} \cdot \mathbf{u} = 0$. $\mathbf{a}$ is a spacelike vector perpendicular to $\mathbf{u}$.

The Four-Current

Baryons are conserved. We define a 4-current $\mathbf{J}$. For an observer $\mathcal{L}_0$ with four-velocity $\mathbf{u}_0$, the local number density is

$$n = \mathbf{J}(\mathbf{u}_0) \tag{16}$$

The 3-current is

$$\mathbf{J} = \mathbf{J}(\mathbf{e}_i)\,\mathbf{e}_i \tag{17}$$

The 4-current for a baryonic fluid

$$\mathbf{J} = n\,\mathbf{u} \tag{18}$$

and the conservation equation is

$$\nabla \cdot \mathbf{J} = 0. \tag{19}$$

Stress-Energy Tensor

The Stress-Energy tensor $\mathbf{T}$ describes the matter and energy content in a spacetime. Given an observer $\mathcal{O}_0$ with four-velocity $\mathbf{u}_0$, the stress-energy tensor satisfies

- the matter and energy density measured by $\mathcal{O}_0$ is

$$E = \mathbf{T}(\mathbf{u}_0, \mathbf{u}_0).$$

- the matter momentum density measured by $\mathcal{O}_0$ is

$$\mathbf{p} = -\frac{1}{c}\mathbf{T}(\mathbf{u}_0, \mathbf{e}_i)\,\mathbf{e}_i$$

  where $\mathbf{e}_i$ is an orthonormal basis of the hyperplane orthogonal to $\mathcal{O}_0$'s worldline.

- the matter stress tensor measured by $\mathcal{O}_0$ is

$$S_{ij} = \mathbf{T}(\mathbf{e}_i, \mathbf{e}_j).$$

Stress-Energy Tensor Example: Dust

Consider a collection of *non-interacting* particles of rest mass m_0. At each point P in $\mathcal{M}$ we can characterize the distribution by a matter density ρ and a three-velocity $\mathbf{u}$.

- Specialize to the rest frame $\mathcal{O}$ of the particles, so $\mathbf{u} = \mathbf{0}$.
- In $\mathcal{O}$ the rest mass density is

$$\rho_0 = m_0 n_0$$

  where n_0 is the number of particles per unit volume.
- Frame $\mathcal{O}'$ moves with velocity $\mathbf{v}$ w.r.t. $\mathcal{O}$. In $\mathcal{O}'$ the number density is $n' = \gamma_v n_0$ and the mass of each particle is $\gamma_v m$. The matter density is

$$\rho' = {\gamma_v}^2 \rho_0.$$

Dust

Dust is characterized by ρ_0 and $\mathbf{u}$. The stress-energy tensor must be constructed from these quantities such as to meet the requirements. The obvious choice is

Dust stress-energy tensor

$$\mathbf{T} = \rho_0 \mathbf{u} \otimes \mathbf{u}, \qquad T^{ab} = \rho_0 u^a u^b.$$

To understand the physical meaning of $\mathbf{T}$, we write out the components in a Cartesian frame $\mathcal{O}'$, where $\mathbf{u} = \gamma_v(c, \vec{v})$.

- $T^{00} = \gamma_v^2 \rho_0 c^2$. The energy density of the particles,
- $T^{0i} = \gamma_v^2 \rho_0 c v^i$. The the energy flux $\times\, c^{-1}$ in the i-th direction,
- $T^{i0} = \gamma_v^2 \rho_0 c v^i$. The the momentum density $\times\, c$ in the i-th direction,
- $T^{ij} = \gamma_v^2 v^i v^j$. The flux of the i-th component of momentum in the j-th direction.

Stress-energy of a fluid

Generalizing from the stress-energy tensor for dust, we expect that the stress-energy tensor for a *general* fluid in a comoving reference frame will have components corresponding to

- T^{00} is the total energy density, including potential energy and thermal kinetic energy.
- T^{0i} although there is no bulk motion, energy could be transmitted by heat conduction.
- T^{ij} momentum flux from random thermal motion. T^{ii} is the isotropic pressure, and T^{ij} $(i \neq j)$ are the viscous stresses in the fluid.

A perfect fluid

A perfect fluid consists of noninteracting particles, and in the rest frame the particles and their velocities are distributed isotropically. A perfect fluid has (i) *no heat conduction*, (ii) *no viscosity*, and (iii) *no shear stresses.* In (a locally flat) rest frame, symmetry requires that

$$\mathbf{T} = \left\| \begin{matrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{matrix} \right\| \tag{20}$$

Perfect fluid stress-energy tensor

$$\begin{aligned} \mathbf{T} &= (\rho + P/c^2)\,\mathbf{u} \otimes \mathbf{u} + P\,\mathbf{g} \\ T^{ab} &= (\rho + P/c^2) u^a u^b + P g^{ab} \end{aligned}$$

The limit $P \to 0$ corresponds to dust.

Fluid Variables

$$
\begin{aligned}
D &= \mathbf{J}(\mathbf{u}_0) && (21)\\
&= m_B n\, \mathbf{u}\cdot\mathbf{u}_0 && (22)\\
D &= \Gamma \rho_0 && (23)
\end{aligned}
$$

$$
\begin{aligned}
E &= \mathbf{T}(\mathbf{u}_0, \mathbf{u}_0) && (24)\\
&= (\rho + P/c^2)(\mathbf{u}\cdot\mathbf{u}_0)^2 + P\,\mathbf{g}(\mathbf{u}_0, \mathbf{u}_0) && (25)\\
E &= \Gamma^2(\rho c^2 + P) + P\,\mathbf{g}(\mathbf{u}_0, \mathbf{u}_0) && (26)
\end{aligned}
$$

FIX THIS. IS LAST TERM MULTIPLIED BY C?

$$
\begin{aligned}
\mathbf{S} &= -\mathbf{T}(\mathbf{u}_0, \mathbf{e}_i)\,\mathbf{e}_i && (27)\\
&= -(\rho + P/c^2)(\mathbf{u}\cdot\mathbf{u}_0)(\mathbf{u}\cdot\mathbf{e}_i) + P\,\mathbf{g}(\mathbf{u}_0, \mathbf{e}_i) && (28)\\
\mathbf{S} &= \Gamma^2(\rho c^2 + P)\mathbf{v} && (29)
\end{aligned}
$$

$$
S_{ij} = \mathbf{T}(\mathbf{e}_i, \mathbf{e}_j) \qquad (30)
$$

A perfect fluid

The Einstein equations are

$$\mathbf{G} = \frac{8\pi G}{c^4}\mathbf{T} \tag{33}$$

Here **G** is the Einstein tensor

$$\mathbf{G} = \mathbf{Ricci} - \frac{1}{2}R\,\mathbf{g} \tag{34}$$

The Einstein tensor satisfies the Bianchi identities

$$\nabla \cdot \mathbf{G} = 0 \tag{35}$$

Similarly for the stress-energy tensor

$$\nabla \cdot \mathbf{T} = 0. \tag{36}$$

This last equation gives the equations of motion.

Fluid Equations of Motion

The fluid equations of motion are

$$\nabla \cdot \mathbf{T} = 0, \qquad \nabla \cdot \mathbf{J} = 0. \tag{37}$$

We will give two formulations of the fluid equations. One in reference to the fluid, and the other applicable to the 3+1 formulations of the Einstein equations. The first is useful in some theoretical work, connecting to the Newtonian limit, etc. The second is used in simulations.
In component form, $\boldsymbol{\nabla} \cdot \mathbf{T} = 0$

$$\nabla_a T^{ab} = u^a u^b \nabla_a(\rho + P/c^2) + (\rho + P/c^2)\left(u^b \nabla_a u^a + u^a \nabla_a u^b\right) + g^{ab}\nabla_a P = 0. \tag{38}$$

Project $\boldsymbol{\nabla} \cdot \mathbf{T} = 0$ along $\mathbf{u}$

$$u_b(\nabla_a T^{ab}) = -u^a \nabla_a(\rho + P/c^2) - (\rho + P/c^2)\nabla_a u^a + u^a \nabla_a P = 0 \tag{39}$$

where we used $u_b \nabla_a u^b = 0$. Using

$$u^a \nabla_a f = \boldsymbol{\nabla}_{\mathbf{u}} f, \qquad \text{and} \qquad \nabla_a u^a = \boldsymbol{\nabla} \cdot \mathbf{u} \tag{40}$$

we get the result

Fluid Equations of Motion

Project $\boldsymbol{\nabla} \cdot \mathbf{T} = 0$ perpendicular to $\mathbf{u}$. The projection operator is

$$\perp = \mathbf{1} + \mathbf{u} \otimes \mathbf{u}, \qquad \perp^a{}_b = \delta^a{}_b + u^a u_b. \tag{42}$$

$$\perp^c{}_b \nabla_a T^{ab} = (\rho + P/c^2) u^a \nabla_a u^c + \nabla^c P + u^c u^a \nabla_a P = 0 \tag{43}$$

This can be written in terms of the four-acceleration $\mathbf{a} = \boldsymbol{\nabla}_{\mathbf{u}} \mathbf{u}$

$$(\rho + P)\mathbf{a} = -\boldsymbol{\nabla} P - (\boldsymbol{\nabla}_{\mathbf{u}} P)\mathbf{u} \tag{44}$$

Newtonian Limit

In special relativity

$$\boldsymbol{\nabla} \cdot \mathbf{T} = 0 \quad \rightarrow \quad \partial_a T^{ab} = 0. \tag{45}$$

The equation projected along **u**

$$\boldsymbol{\nabla} \cdot (\rho \mathbf{u}) + (P/c^2) \boldsymbol{\nabla} \cdot \mathbf{u} = 0 \tag{46}$$

reduces to

$$\boldsymbol{\nabla} \cdot (\rho \mathbf{u}) = 0 \tag{47}$$

or

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{u}) = 0 \tag{48}$$

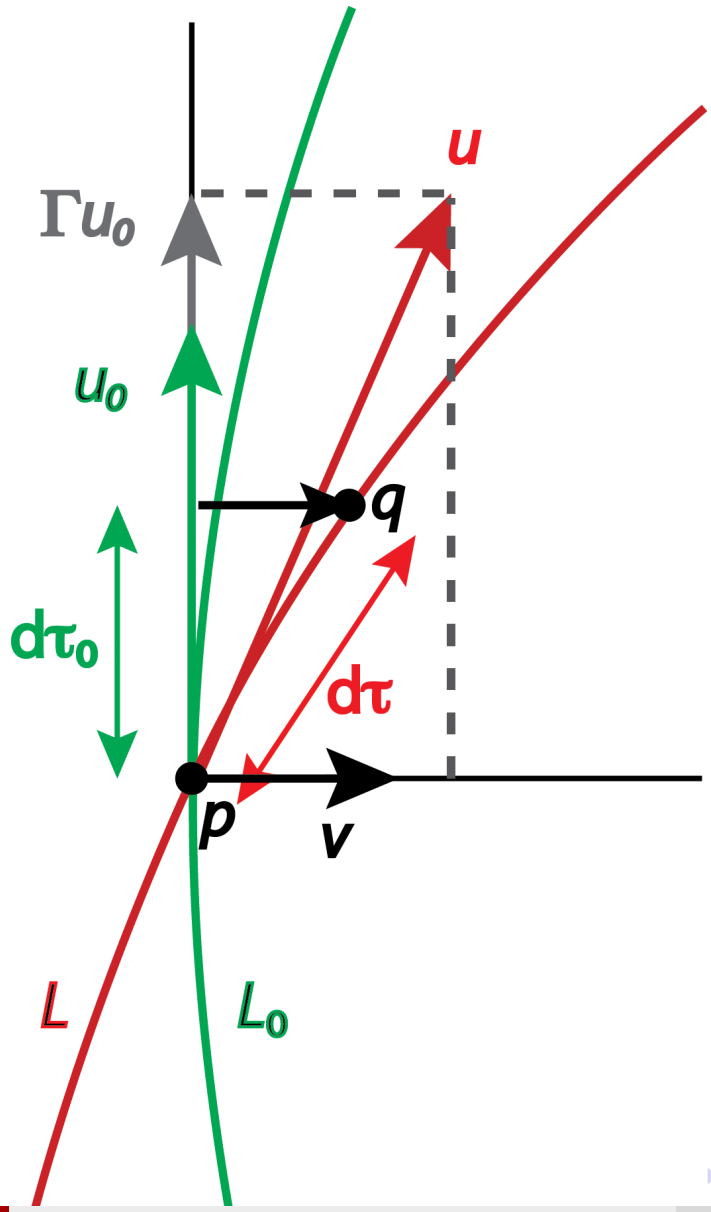
The equation projected perpendicular to **u**

$$\rho \left(\frac{\partial}{\partial t} + \vec{u} \cdot \vec{\nabla} \right) \vec{u} = -\nabla P \tag{49}$$

Then we get

(50)

Fluids in General Relativity



Observers in General Relativity

Consider an observer $\mathcal{O}_0$ with worldline $\mathcal{L}_0$ and proper time τ_0. The set of events in $\mathcal{M}$ that are simultaneous to a given event p are on a hypersurface of $\mathcal{M}$ that is orthogonal to $\mathcal{L}_0$ at p.
Let $\mathcal{O}$ be another observer with world line $\mathcal{L}$ that intersects $\mathcal{L}_0$ at p with proper time τ. After an infinitesimal time interval, $\mathcal{O}$ is at point q at time $\tau + \mathrm{d}\tau$. $\mathcal{O}_0$ is at point r, which is simultaneous with point q, at time $\tau_0 + \mathrm{d}\tau_0$. Then

$$\mathrm{d}\tau_0 = W\,\mathrm{d}\tau, \tag{51}$$

where W is the *Lorentz factor*.

Observers in General Relativity II

Let $\mathcal{O}_0$ and $\mathcal{O}$ have the four-velocities $\mathbf{u}_0$ and $\mathbf{u}$, respectively. Let $\mathrm{d}\mathbf{x}$ be the infinitesimal vector from points r to q, orthogonal to $\mathbf{u}_0$. Then

$$\mathrm{d}\tau\,\mathbf{u} = \mathrm{d}\tau_0\,\mathbf{u} + \mathrm{d}\mathbf{x}. \tag{52}$$

Take the scalar product with $\mathbf{u}_0$.

$$W = -\mathbf{u}_0 \cdot \mathbf{u}, \tag{53}$$

as $\mathbf{u}_0$ and $\mathbf{u}$ are unit vectors, and $\mathbf{u}_0 \cdot \mathrm{d}\mathbf{x} = 0$.

Observers in General Relativity III

The velocity of $\mathcal{O}$ w.r.t. $\mathcal{O}_0$ is

$$\mathbf{v} = \frac{d\mathbf{x}}{d\tau_0}. \tag{54}$$

Then

$$\mathbf{u} = W(\mathbf{u}_0 + \mathbf{v}) \tag{55}$$

with $\mathbf{u}_0 \cdot \mathbf{v} = 0$. Using the normalization conditions $\mathbf{u} \cdot \mathbf{u} = -c^2$ and $\mathbf{u}_0 \cdot \mathbf{u}_0 = -c^2$, gives the Lorentz factor

$$W = \frac{1}{\sqrt{1 - \mathbf{v} \cdot \mathbf{v}/c^2}}. \tag{56}$$

The ADM 3+1 Formulation

Let $\mathcal{O}_0$ be the coordinate observers. The unit timelike normal vector is

$$n_a = (-\alpha, 0, 0, 0), \qquad n^a = \frac{1}{\alpha}(1, -\beta^i) \tag{57}$$

The metric can be written

$$\mathrm{d}s^2 = -(\alpha^2 - \beta_i\beta^i)\,\mathrm{d}t^2 + 2\beta_i\,\mathrm{d}t\mathrm{d}x^i + \gamma_{ij}\,\mathrm{d}x^i\mathrm{d}x^j. \tag{58}$$

α is the lapse function, β^i is the shift vector, and γ_{ij} is the 3-metric induced on the spacelike slices Σ.
From the earlier definitions of the Lorentz factor, we have

$$W = -\mathbf{n} \cdot \mathbf{u} \tag{59}$$

The coordinate velocity

$$\mathbf{v} = \frac{1}{W} \perp \mathbf{u}, \qquad v^a = \frac{1}{W} \perp^a{}_b u^b, \tag{60}$$

Equations of Motion

We define the matter variables

$$E = \mathbf{T}(\mathbf{n}, \mathbf{n}) \qquad \mathbf{S} = -\perp \mathbf{T}(\mathbf{n}, \cdot) \tag{61}$$

In component form

$$\begin{aligned} E &= n_a n_b T^{ab} && (62) \\ &= (\rho_0 + \rho_0 \epsilon + P)\, W^2 - P && (63) \\ S_a &= -\perp_a^c n^b T_{bc} && (64) \\ &= (\rho_0 + \rho_0 \epsilon + P)\, W^2 v_a && (65) \end{aligned}$$

where

$$\begin{aligned} v^i &= \frac{u^i}{W} + \frac{\beta^i}{\alpha} && (66) \\ W &= \frac{1}{\sqrt{1 - v_i v^i}} && (67) \end{aligned}$$

Equations of Motion

The equations of motion are obtained by projecting $\nabla \cdot \mathbf{T}$ along $\mathbf{n}$ and perpindicular to $\mathbf{n}$.

$$n_b \nabla_a T^{ab} = 0 \quad \text{and} \quad \perp^c_b \nabla_a T^{ab} \tag{68}$$

We write these equations in conservation form. Define state vectors of conserved and primitive variables

$$\mathbf{q} = \begin{pmatrix} D \\ S_j \\ \tau \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} \rho_0 \\ v_j \\ \epsilon \end{pmatrix}. \tag{69}$$

Equations of Motion

Then the equations of motion are

$$\frac{1}{\sqrt{-g}}\left(\frac{\partial\sqrt{\gamma}\,\mathbf{q}(\mathbf{w})}{\partial x^0}+\frac{\partial\sqrt{-g}\,\mathbf{F}^i(\mathbf{w})}{\partial x^i}\right)=\mathbf{S}(\mathbf{w}) \tag{70}$$

$$\mathbf{F}^i(\mathbf{w})=\begin{pmatrix} D\left(v^i-\alpha^{-1}\beta^i\right) \\ S_j\left(v^i-\alpha^{-1}\beta^i\right)+P\delta^i{}_j \\ S^i-Dv^i-\alpha^{-1}\beta^i\tau \end{pmatrix} \tag{71}$$

$$\mathbf{S}(\mathbf{w})=\begin{pmatrix} 0 \\ {}^3\Gamma^i{}_{ab}\left(\perp T\right)^a{}_i)+\alpha^{-1}S_a\partial_b\beta^a-\alpha^{-1}\partial_b\alpha E \\ \left(\perp T\right)^{ab}K_{ab}-\alpha^{-1}S^a\partial_a\alpha \end{pmatrix} \tag{72}$$

Quasi Linear Form and the Jacobian Matrix

The equations of motion were written in conservation form. We can also write them as a quasi-linear system

$$\partial_t \mathbf{q} + \mathbf{A} \cdot \nabla \mathbf{q} = \hat{\mathbf{S}}(\mathbf{w}) \tag{73}$$

where $\mathbf{A}$ is the Jacobian matrix

$$\mathbf{A}_{ij} = \frac{\partial F^i}{\partial u^j} \tag{74}$$

The properties of the Jacobian matrix determine the characteristic structure of the solutions. For the general relativistic fluid, the eigenvalues are the characteristic speeds. For fluid moving in the x-direction

$$\lambda_0 = \alpha v^x - \beta^x \qquad \text{(triple)} \tag{75}$$

$$\lambda_\pm = \frac{\alpha}{1 - v^2 c_s^2} \left\{ v^x (1 - c_s^2) \pm c_s \sqrt{(1 - v^2)\left[\gamma^{xx}(1 - v^2 c_s^2) - v^x v^x (1 - c_s^2)\right]} \right\}$$

Characteristic speeds

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The eigenvalues in special relativity reduce to

$$\lambda_0 = v^x \qquad \text{(triple)} \tag{77}$$
$$\lambda_\pm = \frac{1}{1 - v^2 c_s^2} \left\{ v^x (1 - c_s^2) \pm c_s \sqrt{(1 - v^2) \left[(1 - v^2 c_s^2) - v^x v^x (1 - c_s^2) \right]} \right\}$$

and the classical expressions are

$$\lambda_0 = v^x, \qquad \lambda_\pm = v^x \pm c_s^2 \tag{78}$$

Primitive Variable Solver

The conserved variables are

$$D = \rho_0 W, \quad S_i = \rho_0 h W^2 v_i, \quad \tau = \rho_0 h W^2 - P - D. \tag{79}$$

Because the Lorentz factor W combines all components of the velocity, these equations are transcendental, and can not be analytically solved for the primitive variables ρ_0, v^i and ϵ.
These equations must be solved numerically using a root-finding technique such as Newton-Raphson. We first define the unknown $x \equiv \rho_0 h W^2$. Then calculate $S_i S^i$, and solve for W^2 to get

$$W^2 = \frac{x^2}{x^2 - S_i S^i} \tag{80}$$

Note that $\rho_0 = D/W$ and the specific enthalpy is $h = 1 + \epsilon_{\text{int}} + P/\rho_0$.

Primitive Solver Algorithm

Begin with an initial guess for $x = \rho_0 h W^2$

1. Calculate $W(x)$ from

$$W^2(x) = \frac{x^2}{x^2 - S_i S^i} \tag{81}$$

2. Calculate ρ_0

$$\rho_0(x) = \frac{D}{W(x)} \tag{82}$$

3. Calculate $P(x)$ using the definition of h for the ideal gas EOS $P = (\Gamma - 1)\rho_0 \epsilon_{\text{int}}$

$$P(x) = \left(\frac{x}{W^2} - 1\right)\left(\frac{\Gamma - 1}{\Gamma}\right)\rho_0(x) \tag{83}$$

4. Obtain a new guess for x_{new} using the τ equation

$$f = x_{\text{new}} - P(x) - (\tau + D) = 0. \tag{84}$$

Inequalities and the Floor

Numerical methods can sometimes generate unphysical solutions. The following inequalities must be satisified for physical solutions. The fluid equations are well defined when $\rho_0 > 0$. Matter requires $P > 0$ for stability, and relativity requires, of course, $|v| < 1$. In terms of conservative variables, these conditions require

$$D > 0, \qquad \tau > 0. \tag{85}$$

An additional condition can be obtained by requiring that the observed energy density $\mathbf{T}(\mathbf{u}, \mathbf{u})$ is positive in all frames

$$S_i S^i < (\tau + D)^2. \tag{86}$$

This inequality can be sharpened for the ideal fluid EOS with $1 < \Gamma \leq 2$

$$S_i S^i \leq \tau(\tau + 2\rho_0) \tag{87}$$

Exact Solutions

- Riemann Problem
- Tolman-Oppenheimer-Volkoff
- Oppenheimer-Snyder
- Bondi accretion?

Riemann Problem for Relativistic Fluids

The general Riemann problem is an initial value problem for a set of conservation laws

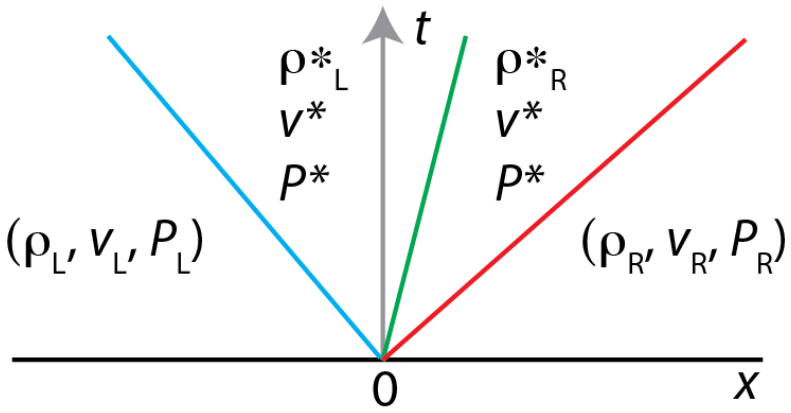
$$\partial_t \mathbf{u} + \partial_x \mathbf{u} = 0 \tag{88}$$

with piece-wise constant intial data

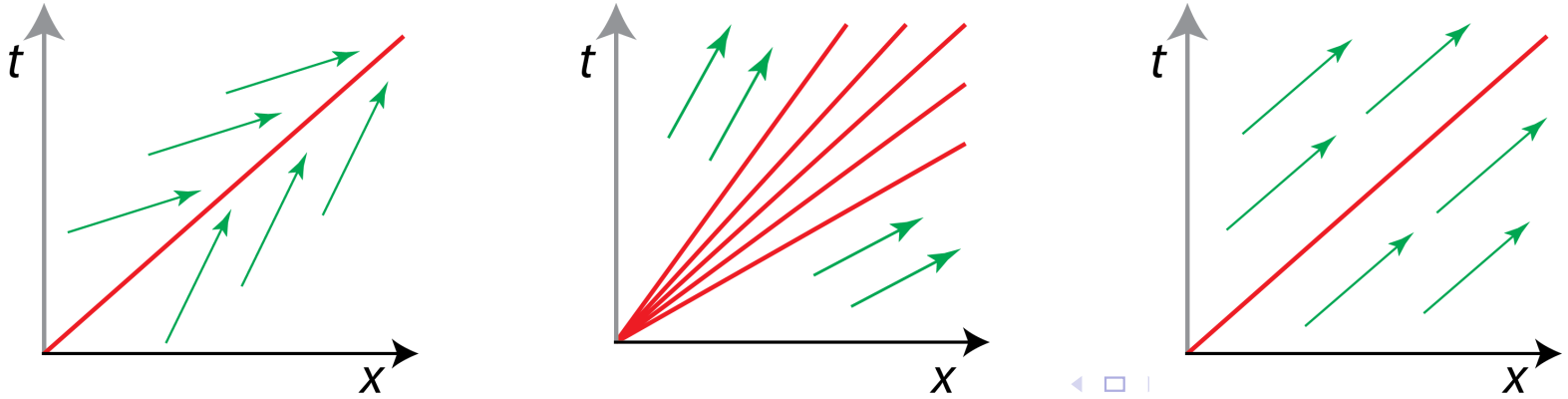
$$\mathbf{u}(x,0) = \begin{cases} \mathbf{u}_\ell & \text{if } x < 0 \\ \mathbf{u}_r & \text{if } x > 0. \end{cases} \tag{89}$$

Here $\mathbf{u}_\ell$ and $\mathbf{u}_r$ are constant state vectors. The Riemann problem has a self-similar solution in the similarity variable $\xi = x/t$.

Riemann Problem for Relativistic Fluids



The solution of the Riemann problem consists of three possible waves. The initial states are in front of the waves, with two new states behind the waves. The wave type is determined by the characteristic speeds on each side of the wave.

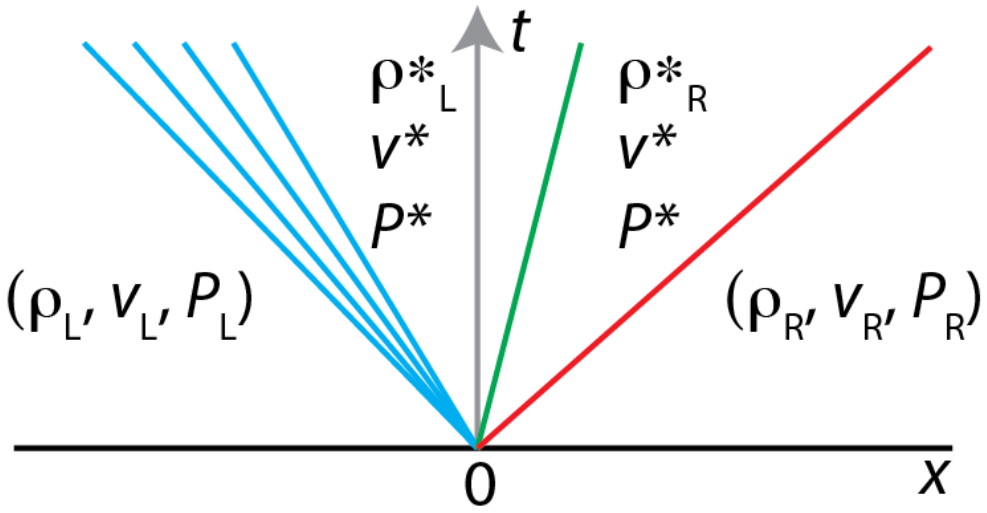


Waves in the Riemann Problem

- **Shock waves:** When characteristics converge, we get compressive waves that are discontinuous (when there is no viscosity). The velocity of shock waves is well-defined by conservation arguments in the Rankine-Hugoniot conditions. Entropy increases behind a shock.
- **Rarefaction waves:** When characteristics diverge, we have rarefaction waves that smoothly connect two constant fluid states. Entropy is constant in rarefaction waves.
- **Contact Discontinuity:** When characteristics are parallel, we have a contact discontinuity.

The solution to the Riemann problem for the Euler equations will always have one contact discontinuity. Only ρ_0 is discontinuous across the contact discontinuity, while the other variables are constant. The other two waves can be either shocks or rarefaction waves, depending on the initial data.

Structure of Riemann Problem

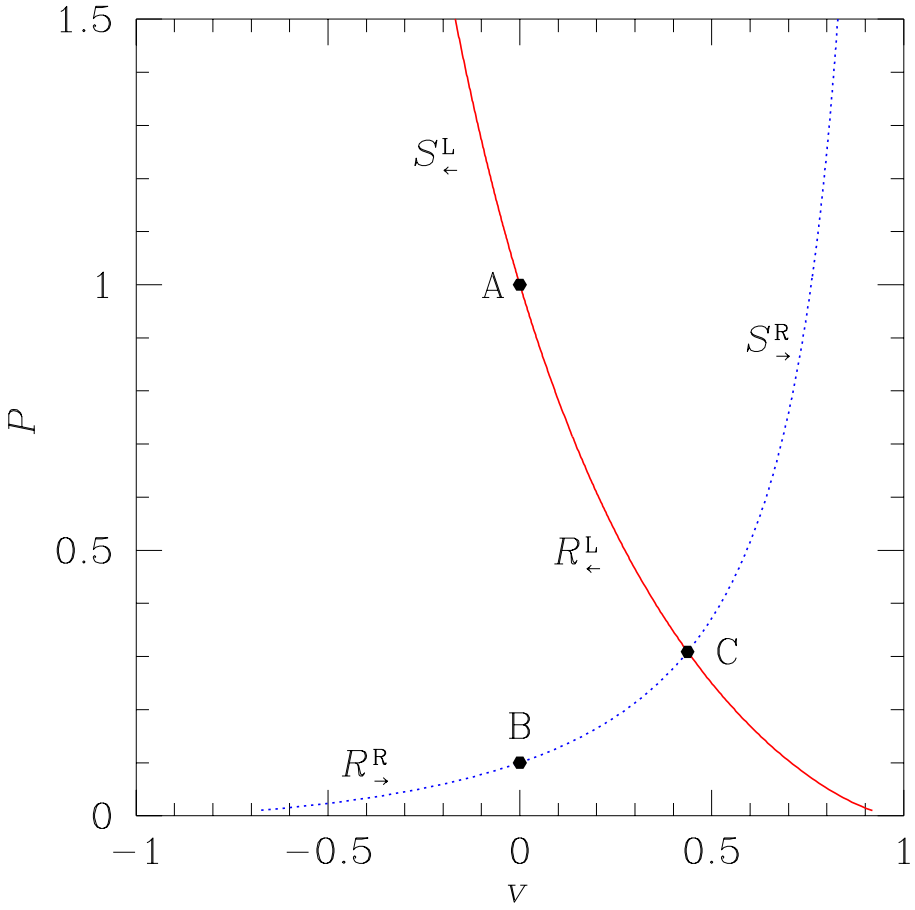


In this diagram, the red line represents a traveling shock wave to the right. (The horizontal axis is a spatial axis, and time is on the vertical.) The green line represents a contact discontinuity, and the cyan lines are the rarefaction fan.

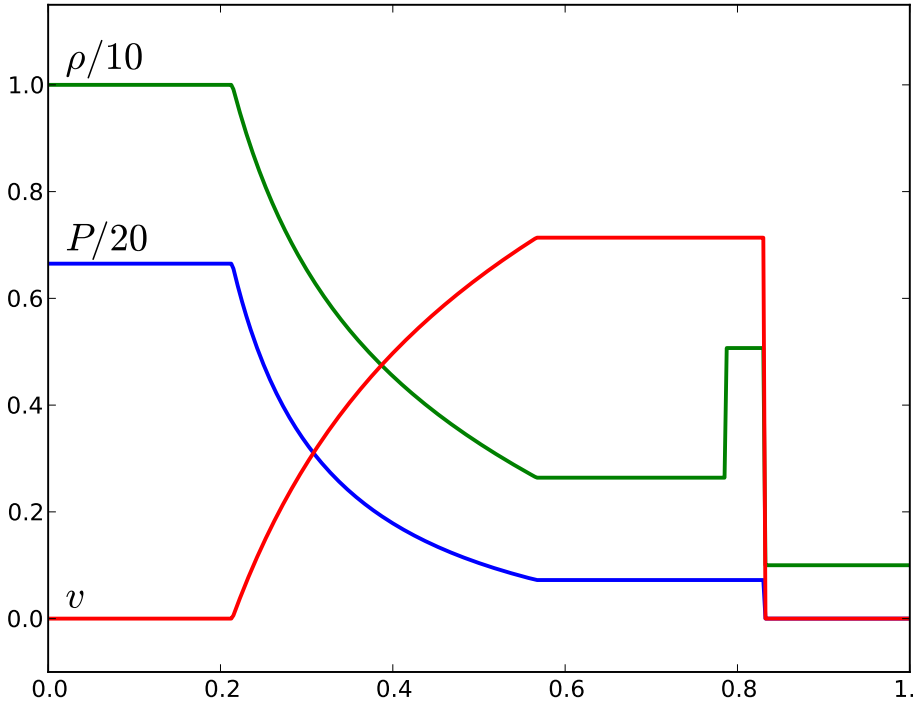
The initial states are given by $\mathbf{w}_L = (\rho_L, v_L, P_L)$ and $\mathbf{w}_R = (\rho_R, v_R, P_R)$. Behind the waves there are two states, $\mathbf{w}_L^*$ and $\mathbf{w}_R^*$, separated by a contact discontinuity. Only ρ is discontinuous across the contact discontinuity.

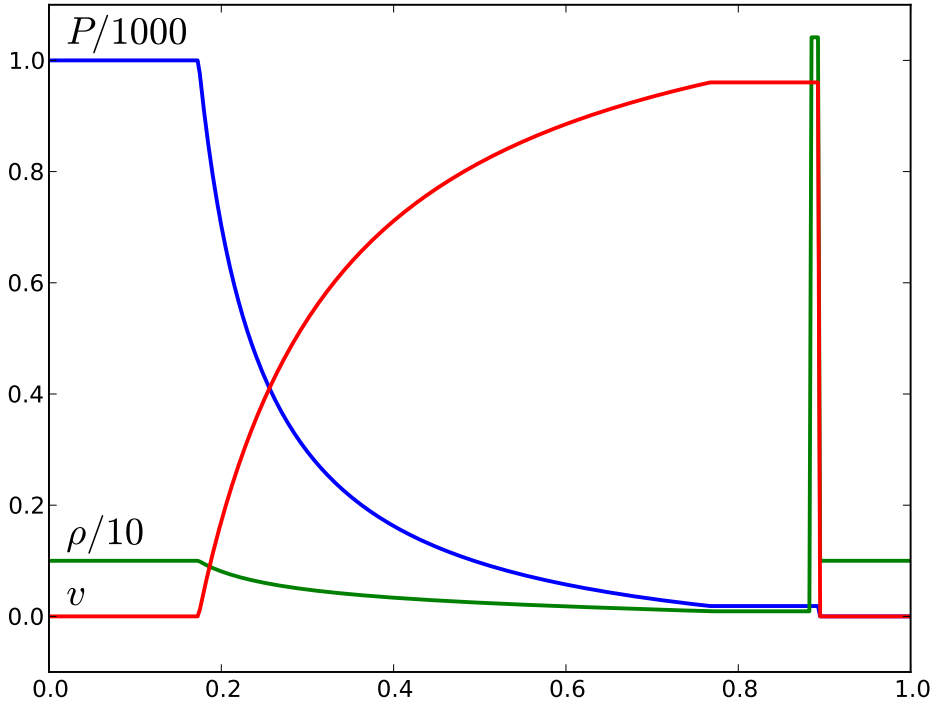
Graphical Solution of the Riemann Problem

In this P-v plot, the initial states are given by the points A and B. Possible left-moving waves that can be connected with point A are given by the red line, which are shocks if $P > P_A$ or rarefaction waves if $P < P_A$. Possible right-moving waves that can be connected with point B are given by the blue dotted line. The solution of the Riemann problem is given by point C, where the two lines cross.

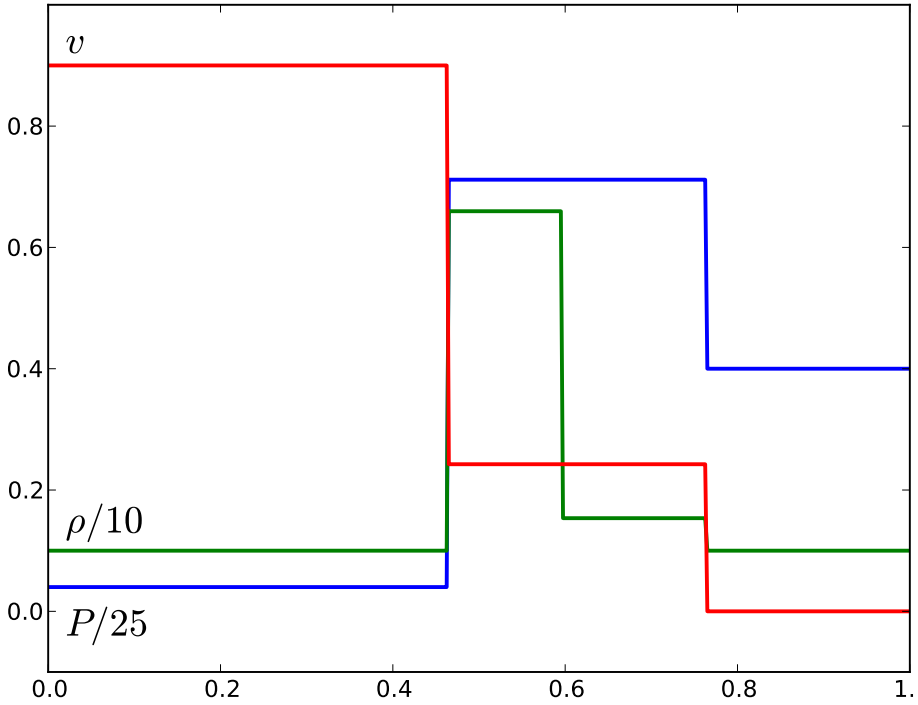


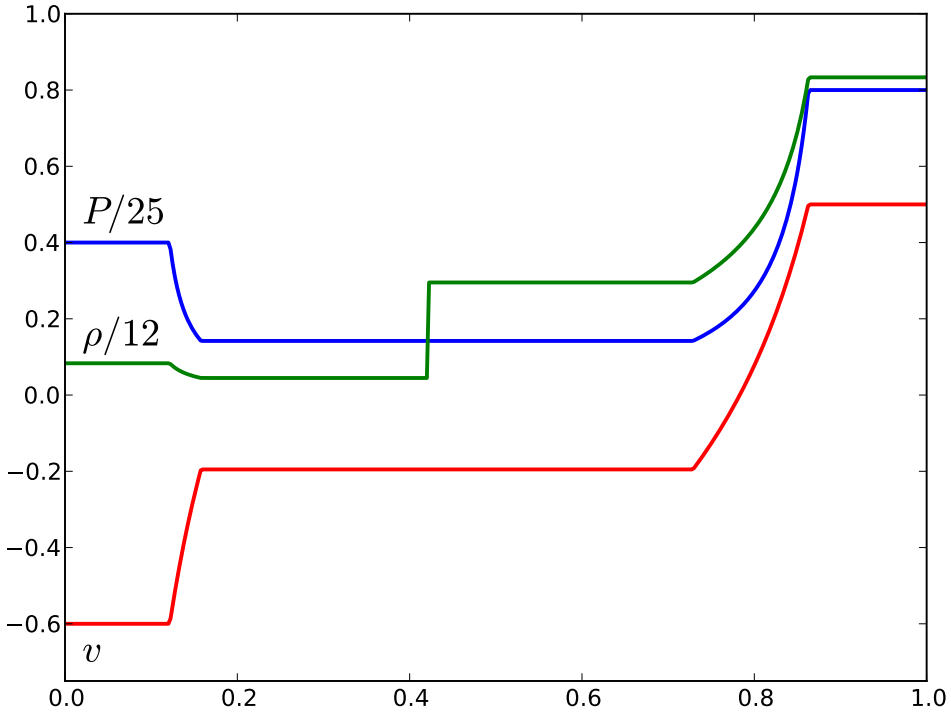
Blast Wave Solutions of the Riemann Problem





More Solutions of the Riemann Problem





References for the Relativistic Riemann Problem

The Riemann problem for relativisitic fluids:

J.M. Martí and E. Müller, "The analytical solution of the Riemann problem in relativistic hydrodynamics," *J. Fluid Mech.* **258** 317–333 (1994).

L. Rezzolla and O. Zanotti, "An improved exact Riemann solver for relativistic hydrodynamics," (2001).

The Riemann problem for relativistic MHD:

B. Giacomazzo and L. Rezzolla, "The exact solution of the Riemann problem in relativistic magnetohydrodynamics," *J. Fluid Mech.* **562** 223–259 (2006).

Interior solutions

- Constant density solution
- TOV solutions

Spherical Star Solutions

We wish to construct a star. The metric in spherical symmetry is

$$\mathrm{d}s^2 = -e^{2\Phi(r)}\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1-\dfrac{2m(r)}{r}} + r^2\,\mathrm{d}\Omega^2 \tag{90}$$

The interior solution must match smoothly to the unique exterior solution

$$\mathrm{d}s^2 = -\left(1-\frac{2M}{r}\right)\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1-\dfrac{2M}{r}} + r^2\,\mathrm{d}\Omega^2 \tag{91}$$

Structure Equations

$$m(r) = \frac{1}{c^2}\int_0^r \mathrm{d}r' 4\pi r'^{\,2}\epsilon(r') \tag{92}$$

$$\mathcal{N}(r) = \int_0^r \mathrm{d}r' 4\pi r'^{\,2} n(r')\left(1-\frac{2Gm}{c^2 r'}\right)^{-1} \tag{93}$$

$$\frac{\mathrm{d}\Phi}{\mathrm{d}r} = \frac{Gm(r)+4\pi G r^3 P/c^2}{r[r-2Gm(r)/c^2]} \tag{94}$$

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{G}{c^2}\frac{(\epsilon+P)}{r^2}\left[m(r)+\frac{4\pi r^3 P}{c^2}\right]\left[1-\frac{2Gm(r)}{c^2 r}\right] \tag{95}$$

- $m(r)$ is a mass function, and the total mass of the star is $M = m(R)$.
- The first equation looks Newtonian, but it is the total gravitational mass the star, including binding energy. Compare the equation for $\mathcal{N}$.
- The total binding energy is $BE = (\mathcal{N}m_0 - M)c^2$.

Solving the TOV Equations

- Set the central density ρ_c for the star, and calculate the central pressure P_c using the EOS. Set $m(0) = 0$ and $\Phi = 0$.
- The TOV equations diverge at $r = 0$, so begin the integration at $r \approx \triangle r$. Set initial data

$$m_0 = m(\triangle r) \approx \triangle m \tag{96}$$

$$P_0 = P(\triangle r) \approx P_c + \triangle P \tag{97}$$

$$\Phi_0 = \Phi(\triangle r) \approx \triangle \Phi \tag{98}$$

- Integrate the TOV equations using the EOS to calculate ρ from P.
- Terminate the integration at $r = R$ where $P = 0$. The star's radius is R and its mass is $M = m(R)$.
- An integration constant can be added to Φ if one desires $g_{00} = 1$ as $r \to \infty$.

Constant Density Solution

If we assume that the star has a constant density, $\rho = \rho_0$, then we can solve the equations exactly.

$$m(r) = \frac{4}{3}\pi r^3 \rho_0 \tag{99}$$

$$P(r) = \rho_0 c^2 \frac{(1 - 2GMr^2/c^2R^3)^{1/2} - (1 - 2GM/c^2R)^{1/2}}{3(1 - 2GM/c^2R)^{1/2} - (1 - 2GMr^2/c^2R^3)^{1/2}} \tag{100}$$

The central pressure is

$$P_c = \rho_0 c^2 \frac{(1 - 2GM/c^2R)^{1/2} - 1}{1 - 3(1 - 2GM/c^2R)^{1/2}} \tag{101}$$

$P_c \to \infty$ if the denominator vanishes, so we require

$$\frac{2GM}{c^2R} < \frac{8}{9} \tag{102}$$

Simple Estimate For Neutron Star Maximum Mass

The constant density solution gives

$$M_{\text{max}} = \frac{8}{27}\left(\frac{3}{4\pi\rho_0}\right)^{1/2} \sim 5.3\left(\frac{4.6\times 10^{14}\text{g/cm}^3}{\rho_0}\right)^{1/2} M_\odot \qquad (103)$$

Largest Precisely Known Neutron Star Masses

PSR J1614-2230	$1.97 \pm 0.04\ M_\odot$	2010
PSR J0348+0432	$2.01 \pm 0.04\ M_\odot$	2013

Maximum Mass for Neutron Stars

The most compact objects result from a soft EOS at low densities and a stiff EOS at high densities. The extreme limit is an EOS like

$$P = \begin{cases} 0, & \epsilon \leq \epsilon_0 \\ \epsilon - \epsilon_0 & \epsilon > \epsilon_0 \end{cases} \tag{104}$$

The maximum mass assuming causality is

$$M_{\text{max}} \simeq 4.09 \left(\frac{\epsilon_s}{\epsilon_0} \right)^{1/2} M_\odot \tag{105}$$

where $\epsilon_s \simeq 150$ MeV is the energy density at nuclear saturation baryon density.

M vs. R and Stability

OLD VERSION: Maximum Mass for Neutron Stars

Simple contraints for neutron star mass

- The density is larger than nuclear density

$$\rho > \rho_{\text{nuc}} \sim 4 \times 10^{14} \text{ g/cm}^3 \tag{106}$$

- The radius is larger than the Schwarschild radius

$$R > \frac{2GM}{c^2} \tag{107}$$

Combining these limits

$$M > \frac{4}{3}\pi R^3 \rho_{\text{nuc}} \tag{108}$$

$$M > \frac{4}{3}\pi \left(\frac{2GM}{c^2}\right)^3 \rho_{\text{nuc}} \tag{109}$$

$$M < \frac{c^3}{G}\sqrt{\frac{3}{32\pi G\rho}} \tag{110}$$

This is about $M < 7M_\odot$.

First Law of Thermodynamics

The particles in each fluid volume are described using thermodynamics. Their random motions and energies are averaged, and described by a few state variables.

In a comoving reference frame $\mathcal{O}$, the First law of thermodynamics is

$$\mathrm{d}E = đQ - đW \tag{111}$$

$đW$ represents a change in mechanical or chemical work, and $đQ$ represents difference in heat energy. $đQ$ and $đW$ are not perfect differentials because they depend on how work is done or heat flows.

The general work term could include

$$đW = P\mathrm{d}V - J\mathrm{d}L - \sigma\mathrm{d}A - \mathbf{E}\cdot\mathrm{d}\mathbf{P} - \mathbf{H}\cdot\mathrm{d}\mathbf{M} - \phi\mathrm{d}e - \sum_j \mu_j \mathrm{d}N_j \tag{112}$$

We simplify and write the First Law as

$$\mathrm{d}E = đQ + P\mathrm{d}V + \sum_j \mu_j \mathrm{d}N_j \tag{113}$$

The Second Law of Thermodynamics

The entropy S plays a central role in thermodynamics. It can be thought of as a measure of disorder in a system. For an isolated system, the entropy is the logarithm of the number of available states. Entropy is related to heat energy

$$\mathrm{d}S \geq \frac{1}{T}đQ, \tag{114}$$

however, for a *reversible* process

$$\mathrm{d}S \equiv \frac{1}{T}đQ \qquad \text{(reversible process only)} \tag{115}$$

The combined First and Second Laws of thermodynamics give the

Thermodynamic Identity

$$\mathrm{d}E = T\,\mathrm{d}S - P\,\mathrm{d}V + \sum_j \mu_j\,\mathrm{d}N_j \tag{116}$$

Intensive and Extensive Variables

When the size of the system changes, some state variables are modified proportionally, while others are constant. The former are *extensive* variables, and the latter are *intensive* variables. Consider two systems, only larger (or smaller) than the other by a factor λ

- Extensive variables
  - Energy, $\tilde{E} = \lambda E$
  - Volume, $\tilde{V} = \lambda V$
  - Entropy, $\tilde{S} = \lambda S$
- Intensive variables
  - Pressure, $\tilde{P} = P$
  - Temperature, $\tilde{T} = T$
  - Chemical Potential, $\tilde{\mu} = \mu$

The Fundamental Equation

We write

$$S(\lambda E, \lambda V, \lambda N_j) = \lambda S(E, V, N_j) \tag{117}$$

Differentiate (λS) w.r.t. λ

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\lambda}(\lambda S) &= \left(\frac{\partial S}{\partial \lambda E}\right)_{V,N_j} \frac{\mathrm{d}}{\mathrm{d}\lambda}(\lambda E) + \left(\frac{\partial S}{\partial \lambda V}\right)_{E,N_j} \frac{\mathrm{d}}{\mathrm{d}\lambda}(\lambda V) \\ &\quad + \sum_j \left(\frac{\partial S}{\partial \lambda N_j}\right)_{E,V,N_{k\neq j}} \frac{\mathrm{d}}{\mathrm{d}\lambda}(\lambda N_j) \end{aligned} \tag{118}$$

Fundamental Equation II

From the Thermodynamic Identity

$$\left(\frac{\partial S}{\partial E}\right)_{V,N_j} = \frac{1}{T}, \tag{119}$$

$$\left(\frac{\partial S}{\partial V}\right)_{E,N_j} = -\frac{P}{T}, \tag{120}$$

$$\left(\frac{\partial S}{\partial N_j}\right)_{E,N_{k\neq j}} = -\frac{\mu_j}{T}. \tag{121}$$

These are the thermal, mechanical, and chemical *Equations of State.* Combined with the previous slide, we obtain

Fundamental Equation of Thermodynamics

$$E = TS + PV + \sum_j \mu_j N_j. \tag{122}$$

Relativistic Considerations

Note that in relativity, e as the total energy also contains the mass energy. We write

$$\epsilon = \sum_j m_j n_j c^2 + \varepsilon_{\text{int}} \tag{123}$$

where m_j is the rest mass of single particle of species j and ε is the internal energy. ε contains the thermal energy, kinetic energy of random, thermal motion, energies of any internal states.
μ_j is the relativistic chemical potential for particles of species j, and it also contains the rest mass energy. We can express this in terms of the classical chemical potential, μ'_j as

$$\mu_j = \mu'_j + m_j. \tag{124}$$

XXX

For a system with a single particle species, we can simplify the equations by using densities. Let $\lambda = 1/V$. Then

$$\tilde{S} = s, \qquad \tilde{V} = 1, \qquad \tilde{N} = n, \qquad \tilde{E} = \epsilon \tag{125}$$

where

- s is the entropy per unit volume
- n is the number density
- ϵ is the total energy per unit volume.

$$\epsilon = \tilde{E}(\tilde{S}, \tilde{V}, \tilde{N}) = \tilde{E}(s, 1, n) = \tilde{E}(s, n). \tag{126}$$

The Thermodynamic Identity becomes

$$\mathrm{d}\tilde{E} = \tilde{T}\,\mathrm{d}\tilde{S} - \tilde{P}\,\mathrm{d}\tilde{V} + \sum_j \tilde{\mu}_j\,\mathrm{d}\tilde{N} = T\,\mathrm{d}s + \sum_j \mu_j\,\mathrm{d}n_j \tag{127}$$

which becomes

$$\mathrm{d}\epsilon = T\,\mathrm{d}s + \sum_j \mu_j\,\mathrm{d}n_j. \tag{128}$$

XXX

We identify

$$T = \left(\frac{\partial \epsilon}{\partial s}\right)_{n_j}, \qquad \mu_j = \left(\frac{\partial \epsilon}{\partial n_j}\right)_{s, n_{k \neq j}} \tag{129}$$

Combining with the Fundamental Equation

$$P = -\epsilon + s\left(\frac{\partial \epsilon}{\partial s}\right)_{n_j} + \sum_j n_j \left(\frac{\partial \epsilon}{\partial n_j}\right)_{s, n_{k \neq j}} \tag{130}$$